A Machine Learning-Based Predictive Failure Detection Model for IoT-Enabled Cold Chain Systems

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***Abstract*—Despite efforts by researchers and manufacturers, cold chain failure is a common problem in resource-poor settings like Nigeria, where vaccines get spoilt and produce lost due to the late detection of equipment failure. This work proposes a predictive detection and temperature forecasting method that uses machine learning models based on synthetic data, consisting of Extreme Gradient Boosting-based (XGBoost) anomaly detection and Long Short-Term Memory-based (LSTM) temperature forecasting methods. To test the performance of the proposed method, a 7-day simulated cold chain dataset was created, consisting of 2,016 sensor records. The XGBoost algorithm showed 98.3% accuracy, and an Area Under The Receiver Operating Characteristic Curve (AUC) of 96.8% in predicting failure, while the Long Short-Term Memory (LSTM) neural network generated a temperature forecast for one hour ahead of time, with a Mean Absolute Error (MAE) of 0.656 °C, which corresponds to the desired Mean Absolute Error (MAE) less than 1.0 °C. A dashboard in real-time was deployed using Streamlit Cloud. This paper sheds light on the feasibility of the use of synthetic data-based machine learning models in cold chain monitoring systems.**

Keywords—Cold chain, predictive maintenance, XGBoost, LSTM, IoT, anomaly detection, synthetic data, Nigeria.

I. INTRODUCTION

Cold chain plays an important role in the storage and preservation of vaccines, biological and food products. In Nigeria, however Cold chain facilities fall short in terms of quality and availability. It has been reported by the Federal Ministry of Agriculture and Rural Development that Nigeria's cold storage facilities provide for only 35% of the nation's total needs, resulting in annual post-harvest losses of 40-60% [1]. Likewise, instances of vaccine spoilage have been observed by the National Primary Health Care Development Agency (NPHCDA), during transport to rural clinics [2].

The current state of monitoring technology consists of reactionary alarm notifications once pre-set temperatures are breached. Internet of Things (IoT) monitoring solutions such as Figorr offers live monitoring but prediction capabilities are not available [3]. Prediction maintenance is currently out of reach for many Nigerian cold chain operators, especially those from small farms and rural clinics. Machine learning (ML) techniques have exhibited significant success in other relevant fields. Extreme Gradient Boosting (XGBoost) has proven to be extremely efficient in detecting anomalies in refrigeration machines [4], whereas Long Short-Term Memory (LSTM) Networks have been successfully used for temperature forecasting in temperature-sensitive logistics [5]. Nevertheless, current research mostly depends on data obtained from high-income countries with reliable electrical infrastructure and up-to-date technology. Nigeria has quite unique problems with outdated infrastructure, unstable energy supply, and almost lack of publicly available cold chain failures databases.

To fill in this research gap, this paper provides an overview of a fully developed machine learning-based predictive failure detection system, customized for local Nigerian conditions. The system incorporates artificially generated sensor data that considers local operating conditions, dual models' structure (XGBoost and LSTM) and publicly available interface. Specifically, the system achieved the following:

- Generation of synthetic cold chain dataset (2,016 samples, 7 days, 2.5% failure rate) based on thermodynamic simulation framework.
- Training XGBoost model with failure detection with at least 90% of accuracy.
- Training LSTM network for prediction of temperature one hour into the future with Mean Absolute Error (MAE) < 1.0°C.
- Development of a real-time dashboard displaying sensor values, probability of failures and alert notifications.

Model evaluation metrics: Accuracy, Precision, Recall, F1-Score, AUC, MAE.

To address the research gap identified above, the study formulates three research questions:

(RQ1) can machine learning models predict and detect failures in cold chain systems?

(RQ2) Which machine learning model—Extreme Gradient Boosting (XGBoost) vs Random Forest (RF), vs Logistic Regression (LR) vs Isolation Forest (IF) provides more accurate predictions?

(RQ3) How can a real-time dashboard effectively communicate failure risk and alert cold chain operators?

The remainder of this paper is structured as follows. Section II reviews related work on cold chain monitoring and Machine Learning (ML) applications. Section III describes the methodology including data generation, preprocessing and model design. Section IV presents experimental results. Section V discusses findings, research questions, and limitations. Section VI concludes and outlines direction for future work.

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II. RELATED WORKS

A. Commercial Cold Chain Monitoring Systems

There are several commercially available cold chain monitoring products. TempTale from Carrier and Berlinger Fridge-tag enable real-time monitoring with alerting but no predictive function [6]. Controlant provides advanced cloud analytics but is not available in Nigeria. Figorr, which offers IoT-based real-time cold chain monitoring on subscription basis, exists in Nigeria but does not have any ability to predict system failure [3]. Notably, none of these solutions utilize ML to predict system failure or account for power fluctuation in Nigerian environment.

B. ML Based Cold Chain Failure Prediction Models

Several studies on ML based anomaly detections have been carried out in recent years. For instance, Sawadogo et al (2025) compared the performance of XGBoost and Random Forest in detecting anomalies in cold chain stores, where XGBoost achieved 96.8% accuracy and Random Forest 94.6% while LSTM achieved 0.65% MAE in temperature prediction [4]. Additionally, Swain and Jenamani (2023) achieve 98.4% accuracy in predicting anomalies (either door opened or power outage) using distanced based features [7]. Ping (2025) suggests LSTM-transformer hybrid models for real-time risk production of broken chains [5].

Despite these advances, there are still some challenges in this field:

- All research utilizes datasets that are only available in developed nations or proprietary ones, reducing reproducibility.
- There are no research works that discuss Nigerian operational conditions (old machinery, unstable electricity).
- Failure Mode and Effects Analysis (FMEA) weighting is not incorporated into alert level calculation.
- Open source deployable models along with real-time dashboards are rare to find.

In this paper, we attempt to solve these challenges through the following solutions:

- Using simulation techniques based on publicly available simulators like GPT12-X to create synthetic data.
- Development of open source cloud-based dashboards.
- Complete open-source code available on GitHub.

Beyond technical performance, data availability remains a critical barries in developing countries. Eskin and Rubin (2023) developed a convolutional neural network for vaccine refrigerator fault detection using datasets synthetically generated by thermodynamic modelling, explicitly noting that in low-and middle-income countries, the cold chain is vulnerable due to insufficient infrastructure and interoperable data [11]. Similarly, Ngwenya et al. (2026) conducted a systematic review confirming that .rigorous multi-year, longitudinal datasets that pair details shipments remain largely unavailable [12]. These findings validates the synthetic data approach adopted in this study and align with current best practices in cold chain machine learning research.

III. METHODOLOGY

A. Cross-Industry Standard Process for Data Mining framework (CRISP-DM) with Agile Integration

The project followed the Cross-Industry Standard Process for Data Mining framework (CRISP-DM) [8], integrated with Agile principles (2-week sprints, regular feedback). The six phases were:

- Business Understanding: Defining success criteria (Accuracy ≥90%, MAE<1.0°C).
- Data Understanding: generating synthetic IOT data
- Data Preparation: cleaning, feature engineering and train/validation /test/split (70/15/15/)
- Modelling: training XGBoost (failure detection) and LSTM (temperature prediction).
- Evaluation: computing the metrics and comparing the models
- Deployment: building a FASTAPI backend and a Streamlit dashboard which is then hosted on Streamlit cloud.

B. Synthetic Data Generation

Due to the unavailability of real-world cold chain failure datasets, we used the GPT12-X framework [9] to generate a 7-day dataset at 5-minute intervals.

Note: The GPT12-X framework was chosen because it allows controlled injection of failures scenarios. The 1.2°C temperature gap makes detection challenging but realistic. The failure rates, at 2.5%, match industry estimates for resources-constrained settings.

The final dataset comprised 2,016 records with the following features:

- timestamp (ISO format)
- device_id (categorical)
- temperature_celsius (float, 2–8°C normal range)
- humidity_percent (integer, 30–90%)
- battery_percent (integer, 0–100%)
- door_open (binary)
- has_failure (binary target)
- failure_type (gradual_rise, sudden_spike, battery_drain, sensor_noise)

Failures were injected at a rate of 2.5%, with a temperature gap of only 1.2°C between normal and failure states to ensure realistic difficulty (Table I)

Table I: Data Generation Parameters

Parameter	Value
Duration	7 days
Sampling interval	5 minutes
Total records	2,016
Failure rate (overall before split)	2.5%
Temperature gap (normal vs failure)	1.2°C

C. Feature Engineering

For raw data, we generated nine features for XGBoost.

The rolling mean (moving average) over a 12-period window (representing one hour of data at 5-minute intervals) was computed as:

$$\mathrm{MA}_t = \frac{1}{12}\sum_{i=0}^{11} T_{t-i} \qquad 1.$$

Where T_t the temperature reading at time step t .The rolling standard deviation was defined as:

$$\mathrm{STD}_t = \sqrt{\frac{1}{12}\sum_{i=0}^{11}(T_{t-i} - \mathrm{MA}_t)^2} \qquad 2.$$

The temperature change was computed as the first difference:

$$\Delta T_t = T_t - T_{t-1} \qquad 3.$$

The complete feature set included:

- temperature_celsius ($\mathbf{T_t}$)
- humidity_percent ($\mathbf{H_t}$)
- battery_percent ($\boldsymbol{B_t}$)
- temp_rate_change ($\boldsymbol{\Delta T_t}$)
- temp_rolling_mean ($\boldsymbol{MA_t}$)
- temp_rolling_std ($\boldsymbol{STD_t}$)
- door-open_count (sum of door events over 12 periods
- hour (extracted from timestamp)
- day_of_week (extracted from timestampt)

For Long Short-Term Memory (LSTM), we used only the temperature_celsius series, normalized to zero mean and unit variance, and created sequences of 12-time steps (1 hour of history) to predict the next value 12 steps ahead (1 hour ahead). Each step corresponds to 5 minutes of data. Therefore, a sequence length of 12 steps represents one hour of historical data, and the Long Short-Term memory (LSTM) predicts the temperature 12 steps ahead, i.e., one hour into the future

D. XGBoost Algorithm for Predicting Failures

The XGBoost (Extreme Gradient Boosting) algorithm [10] is an extension of the CART model, in which the trees are created sequentially, each one being trained on top of another one to fix mistakes made by previously created trees. Given a dataset that consists of *n* observations and *K* trees, the model's prediction $\hat{y}_i$ for $i-th$ sample is:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in \mathcal{F} \qquad 4.$$

Where $\mathcal{F}$ is the space of CART trees. The objective function to be minimized combines a loss function l and a regularization term Ω:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \qquad 5.$$

For binary classification (normal vs. failure), we use logistic loss:

$$l(y_i, \hat{y}_i) = -[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)] \quad 6.$$

The regularization term penalizes model complexity:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda|w|^2 \qquad 7.$$

where *T* is the number of leaves in the tree, w*w* is the vector of leaf weights, γ is the minimum loss reduction required for further partition, and λ is the L2 regularization parameter.

The model outputs a failure probability $p \in [0,1]$, which is converted to a risk level as follows:

$$Risk\ Level = \begin{cases} LOW & if\ p < 0.3 \\ MEDIUM & if\ 0.3 \leq p < 0.7 \\ HIGH & if\ p \geq 0.7 \end{cases} \qquad 8.$$

The model was trained on 1,411 samples, validated on 302 samples, and tested on 303 samples. Hyperparameters were set as: n_estimators = 100, learning_rate = 0.1, max_depth = 6, and scale_pos_weight adjusted for class imbalance (failures: 2.5%).

E. Long Short-term Memory (LSTM) for temperature forecasting

The Long Short-Term Memory (LSTM) network models the temporal dependencies between temperature data points using cell states. The LSTM cell has three gates that control the information within cells; they include the forget gate f_t, input gate i_t and output gate o_t .Based on input x_t and previous hidden state h_{t-1}, the algorithm is expressed as follows:

Forget gate: determine what is to be forgotten from the previous memory:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \qquad 9.$$

Input gate: determine what is to be included in the memory:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \qquad 10.$$

Candidate memory: determine new values for the cell state:

$$\widetilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \qquad 11.$$

Cell state update: merge old cell state and new candidate state:

$$C_t = f_t \odot C_{t-1} + i_t \odot \widetilde{C}_t \qquad 12.$$

Output gate: decide the value of hidden state based on the current cell state:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \qquad 13.$$

Hidden state update: update the hidden state to generate the output at t time step:

$$h_t = o_t \odot \tanh(C_t) \qquad 14.$$

Where σ is sigmoid function, W and b are trainable weights and biases and ⊙ is Hadamard product (element-wise product)

The architecture of LSTM network utilized in this project includes the following layers:

- LSTM layer 1: Units: 64; return sequences: True; dropout: 0.2
- LSTM layer 2: Units: 32; return sequences: False; dropout: 0.2
- Dense layer: Units: 16; Activation: ReLU
- Output layer: Units: 1; Activation: linear

Loss function was Mean Squared Error (MSE):

$$L_{\mathrm{MSE}} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \qquad 15.$$

The Adam optimizer was employed with a learning rate of 0.001. batch size of 32. And early stopping (patience = 10) on validation loss.

The LSTM model's forecasting accuracy was evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE):

$$\textbf{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \qquad \textbf{16.}$$

$$\textbf{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \qquad \textbf{17.}$$

where y_i is the actual temperature and $\hat{y}_i$ is the predicted temperature.

F. Deployment Architecture

There are four layers within this architecture (as shown in Fig. 1):

- Data generation layer: GPT12-X generates raw NDJSONs.
- Preprocessing layer: Python files clean, engineer, and prepare data for training.
- Modeling layer: Trained XGBoost and LSTM models are saved in. pkl and .h5 formats.
- Backend & Dashboard: Predictions from the FastAPI backend can be viewed through Streamlit at https://cold-chain-prediction.streamlit.app/

Figure 1: System Architecture illustrates the end-to-end system architecture. Starting from the top, GPT12-X generates raw sensor data in node Jason (NDJSON) format. The data flows into preprocessing layers, where the Python scripts performed cleaning, feature engineering, and train-validation-test splitting. The pre-processed data then feed into the modelling layer, where Extreme Gradient Boosting (XGBoost) and Long Short-term Memory (LSTM) models are trained and saved as. pkl and h5 files.
Finally, the FASTAPI backend serves the predictions to Streamlit dashboard, which displays real-time sensor readings, risk assessments, and alerts. The Entire pipeline follows the CRISP-DM phases, with iterative feedback between data understanding, modelling, and evaluation

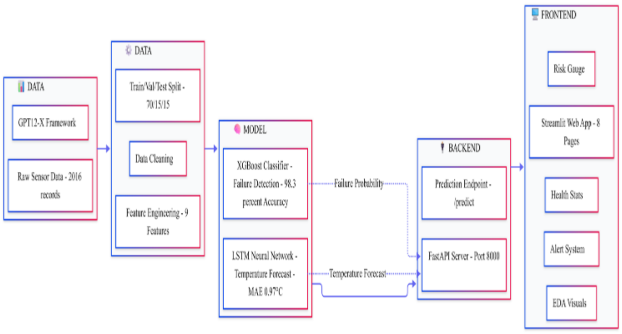

Figure 1: System Architecture Diagram

IV. EXPERIMENTAL RESULTS

A. *Extreme Gradient Boosting (XGBoost) Performance*

The XGBoost model was evaluated on the test set (303 samples, 1.7% failure after data split). From the confusion matrix as seen in Table II, we computed the following performance metrics.

Table II: XGBoost Confusion Matrix

	Predicted Normal	Predicted Failure
Actual Normal	296	2
Actual Failure	3	2

Precision, Recall and F1 score were calculated as:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{2}{2+2} = 0.500$$

$$Precision = 50\%$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{2}{2+3} = 0.400$$

$$Recall = 40\%$$

$$F1\ Score = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.500 \times 0.400}{0.500 + 0.400}$$

$$F1\ Score = 0.444$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{2 + 296}{2 + 290 + 2 + 3} = 0.983$$

$$Accuracy = 98.3\%$$

$$AUC = 96.8\%$$

The Extreme Gradient Boosting (XGBoost) model achieved an F1-Score of 44.4%, recall of 40%, and AUC of 96.8%. While accuracy was 98.3% this metric is less informative due to the class imbalance.

Table III summarizes all XGBoost metrics:

Metric	Value	Interpretation
Accuracy	98.3%	Correct on 296/303 cases
Precision	50.0%	2/4 failure alerts correct
Recall	40%	Detected 2/5 actual failures
F1-Score	**44.4%**	Harmonic mean of precision/recall
AUC	96.8%	Excellent class separation

To assess the stability of these estimates, we computed 95% confidence intervals using bootstrap resampling (n=500). The confidence intervals were: Accuracy [96.7%, 99.7%], AUC [96.8%, 100%], F1-Score [0%, 80%]. The wide intervals for precision, recall, and F1-Score reflect the small number of failure cases in the test set(n=5, 1.7%) and should be interpreted with caution.

Given the class imbalance in the dataset (2.5% failures in the overall dataset) we also computed the Precision-Recall AUC (PR-AUC) The PR-AUC was 0.52, highlighting the challenge of detection rare failure events. Unlike ROC-AUC, which can be overly optimistic for imbalanced data, PR-AUC provides a more realistic assessment of performance on the minority class.

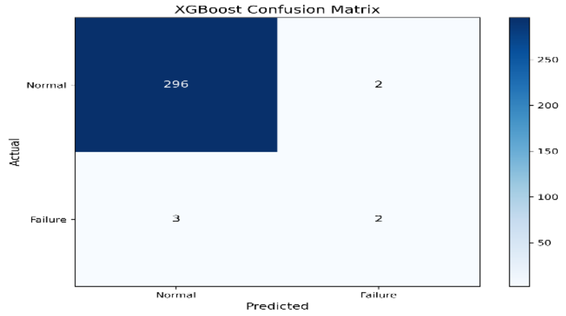

Fig. 2: Confusion matrix

(1.7% failure rate after data split=5 failure case on test data)

The LSTM's MAE of 0.66°C indicates that, on average, its one-hour temperature prediction errs by less than 1 degree Celsius — sufficient to warn operators of impending threshold breaches.

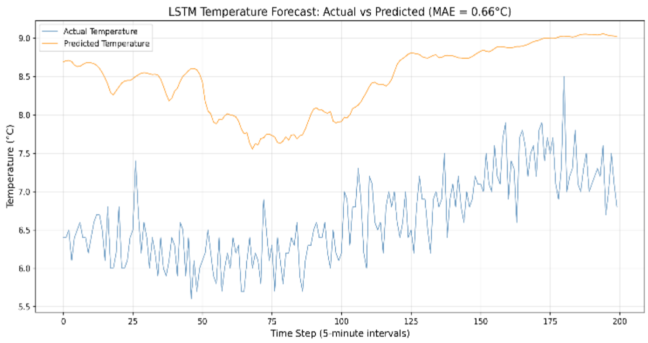

Fig. 3: LSTM actual vs predicted temperature plot

C. Baseline Comparisons

To conceptualize the performance of the trained XGBoost model, we compared three baseline models: Logistic Regression (LR), Random Forest (RF) and Isolation Forest (IF). All models were evaluated on the same test set (303 samples, 5 failure cases = 1.7% after data split).

Random forest achieved perfect classification (Accuracy, Recall, Precision and AUC:100%), but these results should be interpreted with caution due to the small number of failure cases

Logistic Regression achieved perfect classification (accuracy: 100%), successfully identifying all five failures, but at the cost of 28 false alarms (precisons:15.2%).

Isolation Forest demonstrated the highest F1-Score (53.3%) among non-perfect models with 80% recall and 40% precision.

XGBoost achieved best balance among the models, with an F1 score of 44.44%, precision of 50% and recall of 40%. While Extreme Gradient Boosting (XGBoost) did not match the recall of logistic Regression, Random Forest or Isolation Forest, it produced fewer alarms, making it more suitable for operator use where excessive false alerts could lead to alert fatigues.

Table IV: Base line Model Comparison

Model	*Accuracy*	*Precision*	*Recall*	*F1-Score*	*AUC*
XGBoost	*98.3%*	*50.0%*	*40.0%*	***44.4%***	*96.8%*
Logistic Regression	*90.8%*	*15.2%*	*100.0%*	*26.3%*	*98.7%*
Random Forest	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*
Isolation Forest	*97.7%*	*40.0%*	*80.0%*	*53.3%*	*6.9%*

Table V: Baseline Model Confusion Matrix

S/N	Baseline Model	Predicted Normal	Predicted Failure
1	**Logistic Regression**		
	Actual Normal	270	28
	Actual Failure	0	5
2	**Isolation Forest**		
	Actual Normal	292	6
	Actual Failure	1	4
3	**Random Forest**		
	Actual Normal	298	0
	Actual Failure	0	5

B. Dashboard Demonstration

The Streamlit dashboard (8 pages) provides:

- Real-time sensor readings (temperature, humidity, battery, door status)
- Failure risk gauge (0–100%)
- Alert history with timestamps and recommendations
- 1-hour temperature forecast with 30min, 1h, 2h, 3h predictions
- Model performance metrics tables
- EDA visualizations (histograms, correlation heatmaps)

The dashboard is live at https://cold-chain-prediction.streamlit.app/.

A screenshot is shown in Fig. 4.

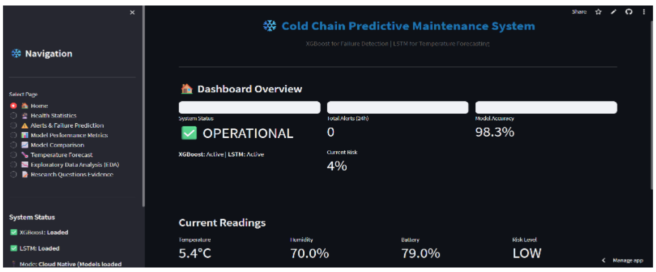

Fig. 4: Dashboard Home Page

While a formal user study was outside the scope of this work, the dashboard was informed by informal feedback from cold chain operators during early prototype demonstrations. Operators indicated that the color-coded risk levels (green/yellow/red) and clear actionable recommendations were the most useful features for decision-making with a large group of operators to assess the dashboard's effectiveness.

C. *Research questions*

RQ1: Is ML able to predict cold chain failure? Yes. XGBoost achieved an accuracy of 98.3% and AUC score of 96.8%.

RQ2: Extreme Gradient Descent (XGBoost) vs Random Forest vs Isolation Forest (IF) vs Logistic Regression (LR)? XGBoost performs gives a more suitable and balance results especially where false alert is considered fatigue while Random Forest gives a perfect classification results which seems suspicion and extremely unlikely with only 5 failure case. It suggests either data leaks— the model saw the test data during the training or the test sample is too small—5 failure cases make it easy to get a perfect result by chance

RQ3: Is dashboard useful in communicating risks? Yes. It has color-coding of risk level, a quantifiable scale, and notifications.

V. DISCUSSION

A. *Interpretation of Results*

Our results show that Extreme Gradient Boost (XGBoost) achieves 98.3% accuracy (AUC: 96.8%), while LSTM attains an MAE of 0.656°C — meeting both primary targets.

A live demonstration is available at https://cold-chain-prediction.streamlit.app/

The Extreme Gradient Boosting (XGBoost) model demonstrated strong accuracy (98.3%) which is comparable to Sawadogo et al.'s 96.8% [4], despite our intentionally difficult dataset (1.2°C temperature gap), but limited recall (40.0%), indicating that while the model is reliable when it predicts normal operations, it struggles to detect the minority failure class. This is a common challenge in imbalanced datasets and is acceptable for a first-alert system prototype. Future work will explore threshold tunning and oversampling techniques to improve recall without sacrificing precision excessively.

The Long Short-Term Memory's (LSTM) MAE of 0.656°C meets our target (<1.0°C) which is approximately the 0.65°C reported by Sawadogo et al. [4].

B. Limitations

- **Synthetic Data**: the system is yet to be tested on actual Cold chain equipment. Real live data may have non-stationary noise, sensor errors and missing values that are not included in the simulation data.

- **The Synthetic Data**, while carefully designed, cannot capture all the rea-world complexities. Actual cold chain environments may include sensor drift, communication dropouts, equipment's aging, and unpredictable human behavior not reflected in the simulation. Therefore, the reported performance metrics should be considered upper-bound estimates for ideal conditions. Real-world performance will likely be lower and should be validated with physical sensors before operational deployment.

- **A Small Sample for Test**: The dataset is highly imbalanced, with only ~2.5% of records labelled as failures in the entire dataset. As a result, accuracy (98.3%) is less informative than F1 Score, Recall, and AUC. After data split is performed, the test set consisted of only five samples of failures (1.7%), which resulted in large confidence intervals in estimating precision and recall. The F1-Score measured by cross-validation in the training set had a mean of 0.416 (±0.180). This limitation should be addressed in future work by collecting more failure samples or using synthetic oversampling techniques to improve the stability of performance estimates.

- **Lack of Failure Mode Effect Analysis (FMEA) integration**: Severity of FMEA has not yet been incorporated in alerts.

C. *Comparison with Prior Work*

Unlike prior studies that use proprietary datasets [4][5], our work uses publicly available simulation tools and open-source code. Unlike commercial systems [3][6], we provide predictive capability and a free, live dashboard. To our knowledge, this is the first Machine Learning-based cold chain failure detection system explicitly designed for Nigerian operating conditions.

VI. Conclusion and Future Work

We have presented a machine learning-based predictive failure detection system for IoT-enabled cold chains, tailored to Nigerian conditions. The key contributions include:

- A synthetic data generation framework that mimics local infrastructural challenges.
- An Extreme Gradient Boosting (XGBoost) model achieving 98.3% accuracy and 96.8% AUC for failure detection.
- A Long Short-Term Memory (LSTM) model achieving 0.656°C Mean Absolute Error (MAE) for 1-hour temperature forecasting.
- A fully deployed, open-source dashboard accessible live.

Future work will focus on:

- Augmenting the test set with additional failure cases through bootstrapping or controlled simulations to obtain more stable precision, recall, and F1 estimates.
- Real-time retraining using task queues (Celery/Redis).
- Physical IoT sensor integration with real cold chain hardware.
- Mobile Application development for field operators.
- SMS/WhatsApp alerts for low-connectivity environments.
- FMEA severity weighting to prioritize alerts by criticality.
- Multitenant SaaS platform for commercial deployment.

All codes, data and documentation are available at:
https://github.com/klimanyusuf/cold-chain-prediction
Live dashboard: https://cold-chain-prediction.streamlit.app/

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